Epileptic Seizure Detection Using Machine Learning: A Comparative Study of Class Imbalance Handling Techniques

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***Abstract*—**Epilepsy is a prevalent neurological disorder affecting over 50 million individuals worldwide, characterized by recurrent seizures arising from abnormal electrical brain activity. Automated detection from electroencephalogram (EEG) signals is essential for timely clinical intervention but is severely challenged by extreme class imbalance, with seizure events comprising less than 2% of continuous EEG recordings. This study presents a systematic comparison of five class imbalance handling strategies on the CHB-MIT Scalp EEG Database. Eight statistical features are extracted from 2-second windows across 23 EEG channels, yielding 184-dimensional feature vectors compressed using Principal Component Analysis (PCA) retaining 120 components. Techniques include Baseline Random Forest, SMOTE with controlled sampling ratio, Balanced Random Forest, SMOTETomek, and XGBoost with scale_pos_weight, with models trained on patients chb01 and chb03 and tested on held-out patient chb05. The Baseline Random Forest with threshold tuning at 0.40 achieved the best F1-score of 0.835 (Precision: 0.956, Recall: 0.741), outperforming all resampling approaches.

Keywords—*EEG, epileptic seizure detection, class imbalance, SMOTE, Random Forest, XGBoost, principal component analysis, CHB-MIT, threshold tuning*

I. Introduction

Epilepsy ranks among the most prevalent chronic neurological disorders worldwide, affecting more than 50 million individuals across all age groups [1]. The condition is defined by recurrent, unprovoked seizures arising from abnormal electrical discharges in the brain. Despite advances in antiepileptic medications, approximately one-third of patients remain pharmacologically refractory, making continuous monitoring and reliable seizure detection an ongoing clinical priority [2].

Electroencephalography (EEG) is the established gold standard for characterising brain electrical activity and forms the primary modality for clinical seizure identification. However, manual interpretation of long-term EEG recordings is time-consuming, subjective, and dependent on trained neurologists—limitations that motivate the development of automated, machine learning-based detection systems [3].

A fundamental challenge is the severe class imbalance inherent in continuous EEG data. Seizure events typically account for less than 2% of total recording time, producing imbalance ratios exceeding 80:1. Standard classifiers trained on such data are biased toward the majority class, producing high overall accuracy but critically insufficient seizure sensitivity. This paper systematically compares five class imbalance handling strategies within a unified pipeline, with additional investigation of decision threshold tuning as a practical post-processing optimisation.

II. Related Work

Shoeb and Guttag [4] introduced the CHB-MIT Scalp EEG Database and proposed a patient-specific SVM-based seizure detector, establishing a widely used benchmark. Subasi and Gursoy [5] demonstrated that PCA-compressed statistical EEG features with SVM yield competitive inter-patient performance. Statistical features including mean, variance, skewness, kurtosis, and RMS have consistently proven discriminative for seizure classification [6].

Chawla et al. [7] introduced SMOTE, which generates synthetic minority samples by linear interpolation; however, its default 1:1 resampling ratio can be too aggressive for extreme imbalances, collapsing classifier precision. Lemaître et al. [8] developed the imbalanced-learn library providing Balanced Random Forest and SMOTETomek implementations. Chen and Guestrin [9] introduced XGBoost, whose scale_pos_weight parameter encodes imbalance directly into the gradient boosting objective. He and Garcia [10] provide a comprehensive review of class imbalance learning, noting no single technique dominates across all domains.

Despite individual evaluations in biomedical contexts, a unified comparative study targeting EEG seizure detection under a cross-patient protocol remains limited. This work fills that gap through a comprehensive evaluation within a consistent experimental pipeline.

III. Dataset and Methodology

A. Dataset

The CHB-MIT Scalp EEG Database [4] comprises long-term scalp EEG recordings from 22 paediatric subjects at Children's Hospital Boston, with 23 channels sampled at 256 Hz and expert-annotated seizure times. Patients chb01 and chb03 form the training set; patient chb05 is the held-out test set, simulating cross-patient generalisation. The training class imbalance ratio is 96.4:1.

B. Preprocessing and Feature Extraction

EEG recordings are segmented into non-overlapping 2-second windows. For each window and each of the 23 channels, eight statistical features are computed: mean, variance, root mean square (RMS), skewness, kurtosis, maximum, minimum, and band power, yielding a 184-dimensional (23×8) feature vector per window. Features are normalised using StandardScaler. PCA retaining 120 components is then applied, fitted exclusively on training data to prevent leakage into the test set.

C. Class Imbalance Handling Techniques

Five strategies are evaluated: (1) Baseline Random Forest (300 estimators, max depth 12) with no correction; (2) SMOTE with sampling_strategy=0.3, raising the minority class to 30% of the majority rather than the default 1:1; (3) Balanced Random Forest, undersampling the majority class within each bootstrap draw; (4) SMOTETomek, combining SMOTE oversampling with Tomek Links removal of borderline majority samples; and (5) XGBoost with scale_pos_weight set to 96.4.

D. Decision Threshold Tuning

For the best-performing model, the classification threshold is varied from 0.10 to 0.50. Lowering the threshold increases seizure recall at the cost of some precision—a clinically relevant tradeoff since missed seizures carry greater risk than false alarms. This is a computationally inexpensive post-processing step requiring no model retraining.

IV. Results and Discussion

A. Technique Comparison

TABLE I shows seizure-class precision, recall, and F1-score for all five techniques on patient chb05. The Baseline RF achieves perfect precision (1.000) but lowest recall (0.638). SMOTE and SMOTETomek both raise recall to 0.828 but reduce precision to 0.632. Balanced RF achieves the highest recall (0.966) at extreme precision cost (0.149), yielding F1=0.259—clinically unsuitable due to excessive false alarms. XGBoost achieves recall 0.862 and F1 0.671. The Baseline RF attains the best overall F1 of 0.779.

TABLE I
Seizure Class Performance Comparison

Technique	Precision	Recall	F1-Score
Baseline RF	1.000	0.638	0.779
SMOTE + RF	0.632	0.828	0.716
Balanced RF	0.149	0.966	0.259
SMOTETomek + RF	0.632	0.828	0.716
XGBoost (SPW)	0.549	0.862	0.671

B. Threshold Tuning Analysis

TABLE II presents threshold tuning on the Baseline RF. Reducing the threshold from 0.50 to 0.40 raises F1 from 0.779 to 0.835, with precision remaining high at 0.956 and recall improving to 0.741. At threshold 0.30, both precision (0.817) and recall (0.845) exceed 0.80, providing the most balanced clinical operating point. These results confirm threshold tuning as a powerful complement to resampling strategies, enabling significant gains without modifying the training process.

TABLE II
Threshold Tuning on Baseline Random Forest

Threshold	Precision	Recall	F1-Score
0.50	1.000	0.638	0.779
0.40	0.956	0.741	**0.835**
0.30	0.817	0.845	0.831
0.20	0.531	0.897	0.667
0.15	0.419	0.931	0.578
0.10	0.300	0.931	0.454

V. Conclusion

This paper presented a systematic comparison of five class imbalance handling techniques for EEG-based epileptic seizure detection using the CHB-MIT dataset under a cross-patient evaluation protocol. The Baseline Random Forest with decision threshold tuning at 0.40 achieved the best F1-score of 0.835 (Precision: 0.956, Recall: 0.741), outperforming all evaluated resampling strategies. Among imbalance methods, SMOTE with a controlled 0.3 sampling ratio and SMOTETomek offered the best recall improvement with acceptable precision.

Two practical conclusions emerge: (i) default 1:1 SMOTE resampling is detrimental for extreme imbalance ratios exceeding 80:1; and (ii) decision threshold tuning on a well-calibrated model outperforms complex oversampling pipelines at no additional training cost. Future work will extend the evaluation to additional CHB-MIT patients, investigate deep learning architectures with focal loss, and explore patient-adaptive threshold selection for real-time deployment.

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